

* Euler Method :-

$$y_{n+1} = y_n + h F(x_n, y_n)$$

Q1. Use Euler method with $h = 0.1$ to find the solution of the equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$

in the range $0 \leq x \leq 0.5$.

Ans $f(x, y) = x^2 + y^2$

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_1 = y_0 + (0.1) f(x_0, y_0)$$

$$y_0 = 0$$

$$y_2 = y_1 + h f(x_1, y_1) = 0 + 0.1 [(0.1)^2 + (0)^2] = 0.001$$

$$y_3 = y_2 + h f(x_2, y_2)$$

x_0	0	y_0	0	$f(x_0, y_0)$	$x_0^2 + y_0^2$	0	$h(x_0^2 + y_0^2)$
x_1	0.1	y_1	$0+0=0$	$f(x_1, y_1)$	$x_1^2 + y_1^2$	0.01	
x_2	0.2	y_2	0.001	$f(x_2, y_2)$	$x_2^2 + y_2^2$	0.04	
x_3	0.3	y_3	0.005	$f(x_3, y_3)$	$x_3^2 + y_3^2$	0.09	
x_4	0.4	y_4	0.014	$f(x_4, y_4)$	$x_4^2 + y_4^2$	0.16	
x_5	0.5	y_5	0.039	$f(x_5, y_5)$	$x_5^2 + y_5^2$	0.25	

$$y_4^{(1)} = 0.795 + \frac{0.2}{3} [0.3937 + 4 \times 0.5689 + 0.7070]$$

$$y_4^{(1)} = 0.795 + 0.06 [0.3937 + 2.2756 + 0.7070]$$

$$y_4^{(1)} = 0.3045 \text{ Ans.}$$

* Newton Raphson method:-

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Q1. Use newton Raphson method find the root of $x^4 - x - 10 = 0$, which is nearer to $x = 2$ correct to 3 places of decimal.

Ans. $f(x) = x^4 - x - 10$

$$f'(x) = 4x^3 - 1$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$x_{n+1} = x_n - \frac{x_n^4 - x_n - 10}{4x_n^3 - 1}$$

$$= \frac{4x_n^4 - x_n^4 + x_n + 10}{4x_n^3 - 1}$$

$$x_{n+1} = \frac{3x_n^4 + 10}{4x_n^3 - 1}$$

First approximation-

$$x_1 = \frac{3x_0^4 + 10}{4x_0^3 - 1} \quad (x_0 = 2) \Rightarrow 1.871$$

Second approximation-

$$x_2 = \frac{3x_1^4 + 10}{4x_1^3 - 1} = \frac{3 \times (1.871)^4 + 10}{4 \times (1.871)^3 - 1} \Rightarrow \frac{46.7634622}{25.1987972}$$

$$x_2 = 1.856$$

Third approximation $x_3 = \frac{3(x_2)^4 + 10}{4(x_2)^3 - 1} = \frac{45.5986183}{24.5737201} \Rightarrow 1.856$

Second and third approximation are same so required root is 1.856.

25/10/2025

* Milne's Method (P-c method) Milne's predictor-corrector method :-

$$y_{n+1}^{(P)} = y_{n-3} + \frac{4h}{3} [2f_{n-2} - f_{n-1} + 2f_n]$$

$$y_{n+1}^{(C)} = y_{n-1} + \frac{h}{3} [f_2 + 4f_3 + f_4^{(P)}]$$

Q1. use milne's P-c method to solve the equation

$$\frac{dy}{dx} = x - y^2 \text{ at } x = 0.8$$

Given $y(0) = 0$

$$y(0.2) = 0.02$$

$$y(0.4) = 0.0795$$

$$y(0.6) = 0.1762$$

x_0	0	y_0	0	$f(x_0, y_0)$	$x_0 - y_0^2$	$0 - 0$	0
x_1	.2	y_1	0.02	$f(x_1, y_1)$	$x_1 - y_1^2$	$.2 - (0.02)^2$	0.1996
x_2	.4	y_2	0.0795	$f(x_2, y_2)$	$x_2 - y_2^2$	$.4 - (0.0795)^2$	0.3937
x_3	.6	y_3	0.1762	$f(x_3, y_3)$	$x_3 - y_3^2$	$.6 - (0.1762)^2$	0.5689
x_4	.8	$y_4^{(P)}$	0.3049	$f(x_4, y_4)$	$x_4 - (y_4^{(P)})^2$	$.8 - (0.3049)^2$	0.7078

$$n=3 \quad y_{3+1}^{(P)} = y_4^{(P)} = y_{3-3} + \frac{4 \times 0.2}{3} [2 \times 0.1996 - 0.3937 + 2 \times 0.5689]$$

$$= 0 + 0.26 [0.3992 - 0.3937 + 1.1378]$$

$$y_4^{(P)} = 0.3049$$

* Bessel's Interpolation Formula :-

$$y_x = \left(\frac{y_0 + y_1}{2} \right) + \left(\frac{u-1}{2} \right) \Delta y_0 + \frac{u(u-1)}{2!} \left[\frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \right] + \frac{\left(u - \frac{1}{2} \right) u(u-1)}{6} \Delta^3 y_{-1}$$

Q1 Use Bessel formula find y_{25} from the following data.

$$y_{20} = 24$$

$$y_{28} = 35$$

$$y_{24} = 32$$

$$y_{32} = 40$$

$$u = \frac{x - x_0}{h} \Rightarrow \frac{25 - 24}{4} = \frac{1}{4} = 0.25$$

$$\frac{32.995313}{33}$$

x	$\frac{x - x_0}{h}$	u	y_u	Δ	Δ^2	Δ^3
20	$\frac{20-24}{4}$	-1			✓	✓
24	$\frac{24-24}{4}$	0	✓	✓	✓	
28	$\frac{28-24}{4}$	1	✓			
32	$\frac{32-24}{4}$	2				

$$\left(u - \frac{1}{2} \right) \cdot u(u-1) \cdot \frac{\left(u - \frac{1}{2} \right) u(u-1)}{6} \text{ \& multiply}$$

* Stirling Formula

$$y_u = y_0 + u \left(\frac{\Delta y_0 + \Delta y_{-1}}{2} \right) + \frac{u^2}{2} \Delta^2 y_{-1} + \frac{u(u^2-1)}{6} \left[\frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right] + \frac{u^2(u^2-1)}{24} \Delta^4 y_{-2}$$

8 use Stirling formula to find y_{28} given

$$y_{20} = 49225$$

$$y_{25} = 48316$$

$$y_{30} = 47236$$

$$y_{35} = 45926$$

$$y_{40} = 44306$$

$$u = \frac{28-30}{5} = \frac{-2}{5} = -0.4$$

Ans

x	$\frac{x-x_0}{h}$	u	y_u	Δy_u	Δ^2	Δ^3	Δ^4
20	$\frac{20-30}{5}$	-2	49225				
25	$\frac{25-30}{5}$	-1	48316				
30	$\frac{30-30}{5}$	0	47236				
35	$\frac{35-30}{5}$	1	45926				
40	$\frac{40-30}{5}$	2	44306				

$$117236 + 478 - 18.4 - 3.8920 + 0.1176$$

$$u = \frac{x - x_0}{h} = \frac{337.5 - 330}{10} = \frac{7.5}{10} = \frac{3}{4}$$

$$\log_{10} 337.5 = y_{3/4} = y_0 + u \Delta y_0 + \frac{u(u-1)}{2} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6} \Delta^3 y_{-1} \\ + \frac{(u-2)(u-1)u(u+1)}{24} \Delta^4 y_{-2} + \frac{(u+2)(u+1)u(u-1)(u-2)}{120} \Delta^5 y_{-2}$$

* Gauss Backward formula :-

2.52821375

$$y_u = y_{3/4} = \log_{10} 337.5 = y_0 + {}^u C_1 \Delta y_{-1} + {}^{u+1} C_2 \Delta^2 y_{-1} \\ + {}^{u+1} C_3 \Delta^3 y_{-2} + {}^{u+2} C_4 \Delta^4 y_{-2} + {}^{u+2} C_5 \Delta^5 y_{-3}$$

$$y_u = y_0 + u \Delta y_{-1} + \frac{u(u-1)}{2} \Delta^2 y_{-1} + \frac{(u+1)u(u-1)}{6} \Delta^3 y_{-2} +$$

$$\frac{(u+2)(u-1)u(u+1)}{24} \Delta^4 y_{-2} + \frac{(u-2)(u-1)u(u+1)(u+2)}{120} \Delta^5 y_{-3}$$

* Gauss Forward Formula :-

Forward Formula:-

$$y_u = y_0 + {}^u C_1 \Delta y_0 + {}^u C_2 \Delta^2 y_{-1} + {}^{u+1} C_3 \Delta^3 y_{-1} + {}^{u+1} C_4 \Delta^4 y_{-2} + \dots$$

$$y_u = y_0 + u \Delta y_0 + \frac{u(u-1)}{2} \Delta^2 y_{-1} + \frac{u(u-1)u(u+1)}{6} \Delta^3 y_{-1} + \frac{(u-2)(u-1)u(u+1)}{24} \Delta^4 y_{-2} + \frac{(u+2)(u+1)u(u-1)u}{120} \Delta^5 y_{-3}$$

Ans 2. 520273

Q. Using Gauss forward formula - find $\log_{10} 337.5$.

x	$\log_{10} x$
337	2.5273
338	2.5292
339	2.5310
340	2.5328
341	2.5346
342	2.5363
343	2.5380
344	2.5397
345	2.5414
346	2.5431
347	2.5447
348	2.5464
349	2.5480
350	2.5497

Ans. x	$u = \frac{x - x_0}{h}$	u	$\log_{10} x = y_u$
310	$\frac{310 - 330}{10}$	-2	2.4913617
320	$\frac{320 - 330}{10}$	-1	2.5051500
fix			
$x_0 = 330$	$\frac{330 - 330}{10}$	0	2.5185139
340	$\frac{340 - 330}{10}$	1	2.5314789
350	$\frac{350 - 330}{10}$	2	2.5440680
360	$\frac{360 - 330}{10}$	3	2.5563025

x	$\log_{10} x$
310	2.4913617
320	2.5051500
330	2.5185139
340	2.5314789
350	2.5440680
360	2.5563025

y_u	$\Delta^2 y_u$	Δ^3	Δ^4
			$\Delta^4 y_{-2}$
$\Delta^2 y_{-1}$	$\Delta^3 y_{-1}$		

$$u \quad \frac{u(u-1)}{2} \quad \frac{(u+1)u(u-1)}{6}$$

• I multiply u

$$\Rightarrow 52 + (-0.3)13 + \frac{(-0.3)(-0.3+1)1}{2} + \frac{(-0.3)(-0.3+1)(-0.3+2)(-4)}{6}$$

$$+ \frac{(-0.3)(-0.3+1)(-0.3+2)(-0.3+3)(-7)}{24} + \frac{(-0.3)(-0.3+1)(-0.3+2)(-0.3+3)(-0.3+4)}{120} \times (-10)$$

$$\Rightarrow 52 - 3.9 - 0.105 + 0.238 + 0.281 + 0.297.$$

$$\Rightarrow \underline{\underline{48.811}}$$

U21-BT527

Q2. find $f(7.5)$

x	$y = f(x)$	∇	∇^2	∇^3	∇^4	∇^5	∇^6	∇^7
1	1							
2	8	7						
3	27	19	12					
4	64	37	18	6				
5	125	61	24	6				
6	216	91	30	6				
7	343	127	36	6				
8	512	169	42	6				

$$(a + nh + uh) = 7.5$$

$$8 + u(10) = 7.5$$

$$u = -0.5$$

$$= 512$$

6/10/2025.

Newton Gregory Interpolation formula:-

Forward Difference table Δ^{delta}

Backward difference table ∇^{neble}

forward table.

year	population	Δ	Δ^2	Δ^3	Δ^4	Δ^5
1931	12	3	2	0	3	-10
1941	15	5	2	3	-7	
1951	20	7	5	-4		
1961	27	12	1			
1971	39	13				
1981	52					

find $f(1934)$

$$1934 = a + uh$$

$$1934 - 1931 = u(10)$$

$$u = 0.3$$

Backward difference table.

year	population	∇	∇^2	∇^3	∇^4	∇^5
1931	12					
1941	15	3				
1951	20	5	2			
1961	27	7	2	0		
1971	39	12	5	3	3	
1981	52	13	1	-4	-7	-10

find $f(1978)$.

$$f(a + nh + uh) = 1978$$

$$1981 + u(10) = 1978$$

$$u = -0.3$$

Newton Gregory backward -

$$f(a + nh + uh) = f(a + nh) + u \nabla f(a + nh) + \frac{u(u+1)}{2!} \nabla^2 f(a + nh) + \frac{u(u+1)(u+2)}{3!} \nabla^3 f(a + nh)$$

$$a + hu = 1934$$

$$1931 + 10u = 1934$$

$$u = 0.3$$

$$12.273$$

$$12.8667 \rightarrow \text{exact}$$

$$F(1934) = \frac{12}{1!} + \frac{0.3}{2!} (3) + \frac{0.3(0.3-1)}{3!} (2) + \frac{0.3(0.3-1)(0.3-2)}{4!} (-10)$$

$$+ \frac{0.3(0.3-1)(0.3-2)(0.3-3)}{5!} (-10) + \frac{0.3(0.3-1)(0.3-2)(0.3-3)(0.3-4)}{6!} (-10)$$

$$\Rightarrow 12 + 0.9 - 0.21 + \frac{0.3(-0.7)(-1.7)}{6} + \frac{0.3(-0.7)(-1.7)(-2.7)}{24} + \frac{0.3(-0.7)(-1.7)(-2.7)(-3.7)}{120}$$

$$\Rightarrow 12 + 0.9 - 0.21 + 0 + 0.120 - 0.291$$

$$\Rightarrow 12.273 \text{ Ans}$$

Q2. Find the cubic polynomial which takes the following data.

$$\begin{array}{c|cccc} x & 0 & 1 & 2 & 3 \\ \hline f(x) & 1 & 0 & 1 & 10 \end{array}$$

$$f(x) = Ax^3 + Bx^2 + Cx + D$$

Ans. forward difference table.

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$a + hu = x$
0	1	-1			$0 + 1u = x$
1	0	1	2		$u = x$
2	1	9	8	+6	
3	10				

$$F(x) = 1 + \frac{x}{1!} (-1) + \frac{x(x-1)}{2!} (2) + \frac{x(x-1)(x-2)}{3!} (6)$$

$$= 1 - x + \frac{x^2 - x}{2} + \frac{x(x^2 - 3x + 2)}{6}$$

$$= 1 - 2x + x^2 + \frac{x^3 - 2x^2 + x}{2} \Rightarrow x^3 - 2x^2 + 1 \text{ Ans}$$

Interpolation with Equal intervals :-

* Newton's forward Interpolations formula :-

	year x	$f(x)$ population y (lakh)
1931 x_0	1931	12
x_1	1941	15
x_2	1951	20
x_3	1961	27
x_4	1971	39
x_5	1981	52

Population $f(x)$

$$h = \text{interval} = 10$$

$$x = 1934$$

$$= a + hu$$

$$\text{Formula } f(a+hu) = f(a) + \frac{u}{1!} \Delta f(a) + \frac{u(u-1)}{2!} \Delta^2 f(a) + \frac{u(u-1)(u-2)}{3!} \Delta^3 f(a)$$

Q1. Find $F(1934)$. find the population of year.
Ans. Δ delta forward table.

x year	$y = f(x) =$ Population	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$	$\Delta^5 f(x)$
$a = 1931$	12					
		3				
1941	15		2			
		5		0		
1951	20		2			
		7		3		
1961	27		5		3	
		12		-4	-7	
1971	39		1			
		13				
1981	52					

Imp. # Lagrange's interpolation Formula:-

$$y = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} y_0$$

$$+ \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} y_1$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} y_2$$

$$+ \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} y_3$$

x_0	y_0
x_1	y_1
x_2	y_2
x_3	y_3

Imp.
Q1. Using Lagrange's formula find the value of y when $x=3$ from the given data.

x	y
$x_0 \leftarrow 0$	$3 \leftarrow y_0$
$x_1 \leftarrow 1$	$4 \leftarrow y_1$
$x_2 \leftarrow 4$	$24 \leftarrow y_2$
$x_3 \leftarrow 5$	$39 \leftarrow y_3$

$$y = \frac{(3-1)(3-4)(3-5)}{(0-1)(0-4)(0-5)} \times 3 + \frac{(3-0)(3-4)(3-5)}{(1-0)(1-4)(1-5)} \times 4$$

$$+ \frac{(3-0)(3-1)(3-5)}{(4-0)(4-1)(4-5)} \times 24 + \frac{(3-0)(3-1)(3-4)}{(5-0)(5-1)(5-4)} \times 39$$

$$y = \frac{2 \times (-1) \times (-2)}{(-1) \times (-4) \times (-5)} \times 3 + \frac{3 \times (-1) \times (-2)}{1 \times (-3) \times (-4)} \times 4 + \frac{3 \times 2 \times (-1)}{4 \times 3 \times (-1)} \times 24$$

$$+ \frac{3 \times 2 \times (-1)}{5 \times 4 \times 1} \times 39$$

$$y = \frac{-12}{20} + \frac{24}{12} + 24 + \left[\frac{-6 \times 39}{20} \right]$$

$$= -0.6 + 2 + 24 - 11.7$$

$$\boxed{y = 13.7} \quad \text{Ans.}$$

Q3 $x + y + z = 6$

$3x + 3y + 4z = 20$

$2x + y + 3z = 13$

Solve by Gauss Elimination and Gauss Jordan's method.

$x=3$
 $y=1$
 $z=2$

$AX = B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 3 & 4 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 20 \\ 13 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 3 & 3 & 4 & 20 \\ 2 & 1 & 3 & 13 \end{array} \right]$$

$R_2 \rightarrow R_2 - 3R_1$ $R_3 \rightarrow R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 0 & 1 & 2 \\ 0 & -1 & 1 & 1 \end{array} \right]$$

$R_1 \rightarrow R_1 - R_3$ $R_2 \rightarrow R_2 - R_3$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 1 & 1 \end{array} \right]$$

$x + 2y = 5$

$y = 1$

$y + z = 1$

$z = 1 + 1$

$z = 2$

$x + 2 = 5$

$x = 3$

By Gauss Jordan's method,

$x = 3$

$y = 1$

$z = 2$

put in question and check the value.

$x + y + z = 6$
 $z = 2$

$-y + z = 1$
 $y = 1$

$x + 3 = 6$

$x = 3$

By Gauss elimination

$x = 3$

$y = 1$

$z = 2$

put in question and check the value.

Augmented matrix $[A : B]$

$$\begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 0 & -3 & 6 & : & 0 \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$R_1 \rightarrow 3R_1 + 2R_2$$

$$\begin{bmatrix} 3 & 0 & 9 & : & 3 \\ 0 & -3 & 6 & : & 0 \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 9R_3, \quad R_2 \rightarrow R_2 - 6R_3$$

$$\begin{bmatrix} 3 & 0 & 0 & : & 21 \\ 0 & -3 & 0 & : & 12 \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$3x_1 = 21 \quad \text{--- (I)}$$

$$-3x_2 = 12 \quad \text{--- (II)}$$

$$x_3 = -2 \quad \text{--- (III)}$$

$$x_3 = -2$$

$$x_2 = -4$$

$$x_1 = 7$$

Check the value by putting in question.

$$+1 - 0 + 2 = 1$$

$$1 = 1 \quad \checkmark$$

$$14 - 4 - 8 = 2$$

$$2 = 2 \quad \checkmark$$

$$21 - 12 - 8 = 1$$

$$1 = 1 \quad \checkmark$$

Done $\Rightarrow$ value same $\Rightarrow$ x, y, z ki value.

diagonal matrix
को के solve करना

↓
Gauss Jordan method.

2023

Solve by Gauss Elimination and Gauss Jordan method.

Q.

$$x_1 + 2x_2 - x_3 = 1$$

$$2x_1 + x_2 + 4x_3 = 2$$

$$3x_1 + 3x_2 + 4x_3 = 1$$

$$AX = B$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 4 \\ 3 & 3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Augmented matrix = $[A : B]$

$$\begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 2 & 1 & 4 & : & 2 \\ 3 & 3 & 4 & : & 1 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 0 & -3 & 6 & : & 0 \\ 0 & -3 & 7 & : & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & -1 & : & 1 \\ 0 & -3 & 6 & : & 0 \\ 0 & 0 & 1 & : & -2 \end{bmatrix}$$

$$x_1 + 2x_2 - x_3 = 1$$

$$-3x_2 + 6x_3 = 0$$

$$x_3 = -2$$

$$-3x_2 = 12$$

$$x_2 = -4$$

$$x_1 - 8 + 2 = 1$$

$$x_1 = 7$$

$$x_1 = 7$$

$$x_2 = -4$$

$$x_3 = -2$$

Check the value by putting in question.

4/10/2025

Gauss Elimination Method :- (upper triangular matrix बनाना)

Q1. Solve $x + 4y + 9z = 16$

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$AX = B$$

$$\begin{bmatrix} 1 & 4 & 9 \\ 2 & 1 & 1 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 16 \\ 10 \\ 18 \end{bmatrix}$$

Augmented matrix $[A:B]$

$$\begin{bmatrix} 1 & 4 & 9 & : & 16 \\ \textcircled{2} & 1 & 1 & : & 10 \\ \textcircled{3} & 2 & 3 & : & 18 \end{bmatrix} \begin{matrix} \rightarrow R_1 \\ \rightarrow R_2 \\ \rightarrow R_3 \end{matrix}$$

$$R_2 \rightarrow R_2 - 2R_1 \quad R_3 \rightarrow R_3 - 3R_1$$

$$\begin{bmatrix} 1 & 4 & 9 & : & 16 \\ 0 & -7 & -17 & : & -22 \\ 0 & -10 & -24 & : & -30 \end{bmatrix}$$

$$R_3 \rightarrow 7R_3 - 10R_2$$

$$\begin{bmatrix} 1 & 4 & 9 & : & 16 \\ 0 & -7 & -17 & : & -22 \\ 0 & 0 & 2 & : & 10 \end{bmatrix}$$

$$x + 4y + 9z = 16 \quad \text{--- (i)}$$

$$-7y - 17z = -22 \quad \text{--- (ii)}$$

$$2z = 10 \quad \text{--- (iii)}$$

$$z = 5$$

$$y = -9$$

$$x = 7$$

Check the value

$$7 + 4(-9) + 45 = 16$$

$$16 = 16 \quad \underline{\underline{\text{Ans.}}}$$

Q1. $\frac{dy}{dx} = x + y^2$ $y(0) = 1$ $y = ?$ $x = 0.2$ by RK Method.

$$y_{n+1} = y_n + K$$

$$y_1 = y_0 + K$$

$$K_1 = h f(x_0, y_0)$$

$$= 0.2 (0 + 1) = 0.2$$

$$K_2 = h f\left[x_0 + \frac{h}{2}, \left(y_0 + \frac{K_1}{2}\right)^2\right]$$

$$= 0.2 \left[0 + \frac{0.2}{2} + \left(1 + \frac{0.2}{2}\right)^2 \right]$$

$$K_2 = 0.2 [0.1 + 1.21] = 0.262$$

$$K_3 = h f\left[x_0 + \frac{h}{2}, \left(y_0 + K_2/2\right)^2\right]$$

$$= 0.2 \left[0 + \frac{0.2}{2} + \left(1 + \frac{0.262}{2}\right)^2 \right]$$

$$= 0.2 [0.1 + (1.131)^2] = (1.379) 0.2 \Rightarrow 0.275$$

$$K_4 = h f[x_0 + h, (y_0 + K_3)^2]$$

$$= 0.2 [0 + 0.2 + (1 + 0.275)^2]$$

$$= 0.2 [0.2 + 1.625]$$

$$= 0.2 (1.825) \Rightarrow 0.365$$

$$K = \frac{K_1 + 2K_2 + 2K_3 + K_4}{6}$$

$$= \frac{0.2 + 0.524 + 0.55 + 0.365}{6} \Rightarrow 0.2731$$

$$y_1 = y_0 + K$$

$$y_1 = 1 + K$$

$$y_1 = 1 + 0.2731$$

$$y_1 = 1.2731$$

Q1. $\frac{dy}{dx} = x + y^2$, $x = 0$, $y(0) = 1$, $x = 0.2$ find y .

Q2. $\frac{dy}{dx} = x + y$, $y = 1$ when $x = 0$.
Find y when $x = 0.2$ by RK method.

Ans. $y_{n+1} = y_n + k$

$y_1 = y_0 + k$

$y_1 = 1 + k$

$y = 1 + 0.2428$

$y = 1.2428$ Ans.

y_0	1	x_0	0
y_1	$y_0 + k$	x_1	0.2
	1 + k		
	1.2428		

$k_1 = h f(x_0, y_0)$

1.2428 Ans

$k_1 = 0.2 (x_0 + y_0)$

$k_1 = 0.2 [0 + 1] = 0.2$

$k_2 = h f(x_0 + h/2, y_0 + k_1/2)$

$k_2 = 0.2 [0 + 0.1 + 1 + 0.1]$

$k_2 = 0.2 [0.1 + 1.1] = 0.24$

$k_3 = h f(x_0 + h/2, y_0 + k_2/2)$

$k_3 = 0.2 [0 + 0.1 + 1 + 0.12]$

$k_3 = 0.2 [0.1 + 1.12] = 0.244$

$k_4 = h f(x_0 + h, y_0 + k_3)$

$k_4 = 0.2 [0 + 0.2 + 1 + 0.244]$

$k_4 = 0.2 [1.444] = 0.2888$

$k = \frac{k_1 + 2k_2 + 2k_3 + k_4}{6}$

$k = \frac{0.2 + 0.48 + 0.488 + 0.2888}{6}$

$k = 0.2428$

Trigonometry $\rightarrow$ degree
exponential $\rightarrow$ Radian

x_0	0	y_0	$e^{-x_0^2}$	e^{-0^2}	1
x_1	0.1	y_1	$e^{-x_1^2}$	$e^{-.1^2}$	0.99
x_2	0.2	y_2	$e^{-x_2^2}$	$e^{-.2^2}$	0.9608
x_3	0.3	y_3	$e^{-x_3^2}$	$e^{-.3^2}$	0.9139
x_4	0.4	y_4	$e^{-x_4^2}$	$e^{-.4^2}$	0.8521
x_5	0.5	y_5	$e^{-x_5^2}$	$e^{-.5^2}$	0.7788
x_6	0.6	y_6	$e^{-x_6^2}$	$e^{-.6^2}$	0.6977

$$= \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{0.1}{3} [(1 + 0.6977) + 4(0.99 + 0.9139 + 0.7788) + 2(0.9608 + 0.8521)]$$

$$= \frac{0.1}{3} [1.6977 + 4(2.6827) + 2(1.8135)]$$

$$= \frac{0.1}{3} [1.6977 + 10.7308 + 3.627]$$

$$= 0.5351 \text{ Ans.}$$

V.V Imp

RK Method (Runge Kutta method of 4th order) :-

$$\begin{aligned} y_{n+1} &= y_n + k \\ k &= \frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \end{aligned}$$

$$k_1 = h f(x_n, y_n)$$

$$k_2 = h f\left[\left(x_n + \frac{h}{2}\right), y_n + \frac{k_1}{2}\right]$$

$$k_3 = h f\left[\left(x_n + \frac{h}{2}\right), y_n + \frac{k_2}{2}\right]$$

$$k_4 = h f[x_n + h, y_n + k_3]$$

$$Q_1. \int_0^6 \frac{dx}{1+x^2}$$

$$= \frac{3 \times 1}{8} [(1 + 0.027) + 3(0.5 + 0.2 + 0.0385 + 0.0588) + 2(0.1)]$$

$$= \frac{3}{8} [1.027 + 2.3919 + 0.2]$$

$$= 1.3570 \text{ Ans.}$$

Weddle's Rule:-

$$\int_a^b f(x) dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

$$Q. \int_0^6 \frac{dx}{1+x^2}$$

$$= \frac{3}{10} [1 + 5(0.5) + 0.2 + 6(0.1) + 0.0588 + 5(0.0385) + 0.027]$$

$$= \frac{3}{10} [1 + 2.5 + 0.2 + 0.6 + 0.0588 + 0.1925 + 0.027]$$

$$= \frac{3}{10} [4.5783]$$

$$= 1.3735.$$

Q1. $\int_0^{0.6} e^{-x^2} dx$ by Simpson's $\frac{1}{3}$ rule.

$$h = \frac{b-a}{n} = \frac{0.6-0}{6} = 0.1$$

$$h = 0.1$$

$$n = 6$$

$$= \frac{1}{2} [1.027 + 1.7946]$$

$$= \frac{1}{2} [2.8216]$$

$$\Rightarrow 1.4108.$$

Actual value -

$$\int_0^6 \frac{dx}{1+x^2} = [\tan^{-1}x]_0^6 = \tan^{-1}6 - \tan^{-1}0 \text{ (Radian)}$$

$$= 1.4056 - 0$$

$$\Rightarrow 1.4056. \text{ Ans.}$$

Simpson $\frac{1}{3}$ rd rule :-

$$\int_a^b f(x) dx = \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

Q. = $\int_0^1 \frac{dx}{1+x^2}$ by Simpson $\frac{1}{3}$ rd rule.

$$= \frac{h}{3} [(y_0 + y_6) + 4(y_1 + y_3 + y_5) + 2(y_2 + y_4)]$$

$$= \frac{1}{3} [(1 + 0.027) + 4(0.5 + 0.1 + 0.385) + 2(0.2 + 0.588)]$$

$$= \frac{1}{3} [1.027 + 2.554 + 0.5176]$$

$$= \frac{1}{3} [4.0986]$$

$$= 1.3662.$$

Simpson $\frac{3}{8}$ Rule :-

$$\int_a^b f(x) dx = \frac{3h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)]$$

27/09/2025.

Unit-2.

Numerical Integration

vJmk.

Trapezoidal Rule:-

$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{h}{2} \left[(y_0 + y_n) + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) \right]$$

Petroleum
2023

Q1. Use the trapezoidal rule with $n=6$ to approximate $\int_0^{\pi} \sin^2 x dx$.

Q2. Solve $\int_0^6 \frac{dx}{1+x^2}$ by trapezoidal rule.

Ans $h = \frac{b-a}{n} = \frac{6-0}{6} = \frac{6}{6} = 1.$

$h=1, n=6.$

x_0	0	$y_0 = f(x_0) = \frac{1}{1+x^2} = \frac{1}{1+0} = 1$
x_1	1	$y_1 = f(x_1) = \frac{1}{1+1^2} = \frac{1}{2} = 0.5$
x_2	2	$y_2 = f(x_2) = \frac{1}{1+2^2} = \frac{1}{5} = 0.2$
x_3	3	$y_3 = f(x_3) = \frac{1}{1+3^2} = \frac{1}{10} = 0.1$
x_4	4	$y_4 = f(x_4) = \frac{1}{1+4^2} = \frac{1}{17} = 0.0588$
x_5	5	$y_5 = f(x_5) = \frac{1}{1+5^2} = \frac{1}{26} = 0.0385$
x_6	6	$y_6 = f(x_6) = \frac{1}{1+6^2} = \frac{1}{37} = 0.027$

$$= \frac{h}{2} \left[(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5) \right]$$

$$= \frac{1}{2} \left[(1 + 0.027) + 2[0.5 + 0.2 + 0.1 + 0.0588 + 0.0385] \right]$$

$$= \frac{1}{2} \left[1.027 + 2[0.8973] \right]$$